

Using Confidence Probes to Estimate Output Confidence in Language Models

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Abstract

We propose confidence probes for estimating output confidence in language models. Our codebase is available [here](#).

1 Problem Statement

In the last few years, language models have gained unrivaled mainstream exposure, becoming the face of Artificial Intelligence. With this widespread adoption, language models have become deeply integrated in critical workflows spanning numerous domains. These include tasks such as factual question answering, document summarization, and medical/legal/technical guidance. The information these models provide can carry significant weight, so as reliance grows, so does the potential impact of their outputs on critical decision-making processes.

However, despite their impressive capabilities, language models are prone to a problem known as hallucination. This phenomenon occurs when models generate outputs that appear confident and authoritative but are factually incorrect, fabricated, or unsupported by evidence. Hallucinations can take many forms, including invented citations, creating their own history, and contradicting statements within the same response. Most importantly, these outputs are often indistinguishable from accurate information, as models typically do not disclose uncertainty or flag potentially unreliable content.

The consequences of language model hallucinations are extensive. Users who trust model outputs without verification may act on false information, leading to misinformed decisions or tangible harm in real-world applications. Thus, both users and developers of these systems need to

understand when and why a model's output may be incorrect, making hallucination detection one of the most crucial challenges in NLP research.

2 What you proposed vs. what you accomplished

In our proposal, we intended to create a selective question-answering pipeline that used a lightweight confidence probe to decide whether to answer, abstain, or choose the best of multiple sampled answers at inference time. We would achieve this by extracting internal model signals and training a small model to output a calibrated probability of correctness. Then, we would apply a threshold to drive the decision policy, and use n -best ranking if the probability was less than the threshold. Original plans were to focus on closed-book QA, primarily with the TriviaQA dataset and several open-weight LLMs.

Throughout the course of this project, we have:

- Trained various types of probes to act as correctness classifiers using specific features from the language model.
- Added multiple datasets (TriviaQA, SquadV2, MMLU, HotpotQA, and GSM8K) and carried out ablation studies per-dataset to find failure cases, as well as assessed cross-domain generalizability for confidence probes.
- Used our pipeline on multiple different model sizes for the same model family (Qwen2.5) to assess whether these probes are generalizable within the same model family.

As mentioned earlier, we also intended to run experiments with techniques such as n -best

ranking during inference (choosing the output with the highest confidence rating out of n rollouts for a given prompt), but this proved to be too computationally intensive for us to complete.

Additionally, we also initially wanted to work with a reasoning model (specifically GPT-OSS), but this proved too finicky to handle as it required very delicate prompting and parsing, so we did not pursue this.

3 Related work

3.1 Hallucination

As shown by Lee et al. (2023), neural language models are highly prone to hallucination, producing outputs that are ungrounded in the input or training data.

3.1.1 Models Know What They Know

Kadavath et al. (2022) show that language models can estimate both the probability that an answer is correct ($P(Truth)$) and whether they know the answer ($P(IK)$). Their results demonstrate that these estimates are reasonably calibrated and generalize to out-of-distribution settings.

3.1.2 Verbalized Confidence Scoring

Verbalized confidence methods prompt models to output confidence scores alongside answers (Yang et al., 2024). While larger models show improved correlation between confidence and accuracy, substantial overconfidence remains, and performance is sensitive to what prompts are used. This makes verbalized confidence unreliable.

3.1.3 Fidelity

Zhang et al. (2024) introduce a prompting-based method to rank MCQ answers by adding a counterfactual option (“All other answers are wrong”). The approach performs well on multiple-choice tasks but does not naturally extend to open-ended question answering.

3.1.4 Stable Explanations

Becker and Soatto (2024) propose assigning confidence based on the stability of reasoning traces across multiple generations. Confidence is high only when explanations are both consistent and relevant. While effective, this black-box approach is computationally expensive and better suited to MCQ-style settings.

3.1.5 RLHF for Overconfidence

Leng et al. (2025) introduce PPO-based RLHF variants that explicitly incorporate confidence into reward modeling. These methods reduce calibration error while maintaining accuracy, but still rely on verbalized confidence and prompt sensitivity.

3.1.6 INSIDE

Chen et al. (2024) detect hallucinations using variance in internal sentence embeddings across multiple sampled generations, quantified via an EigenScore. Test-time feature clipping further reduces overconfident hallucinations. However, the method is computationally intensive due to multiple generations and eigenvalue computations.

3.1.7 RLCR

Damani et al. (2025) optimize both correctness and calibration using a Brier-score-based RL objective. While this improves ECE, the models can still exhibit confident errors.

3.2 Dynamic Information Retrieval

Su et al. (2024) propose triggering retrieval dynamically based on token-level uncertainty, combining entropy, attention influence, and semantic filtering. Retrieval is invoked only when uncertainty exceeds a threshold, improving efficiency over static RAG pipelines.

3.3 Knowing Before Saying

Afzal et al. (2025) show that hidden states can predict the success of zero-shot chain-of-thought reasoning before generation begins. A lightweight probe enables early stopping, preserving reasoning benefits while reducing unnecessary computation.

3.4 MICE

Subramani et al. (2025) introduce Model-Internal Confidence Estimators (MICE), which compare intermediate-layer decodes with the final output to estimate correctness. A learned classifier combines these signals to improve calibration and has been extended to tool-using models.

4 Dataset

For training the probes, we construct a dataset of internal language-model statistics collected on five benchmark datasets spanning a range of tasks and difficulties: TriviaQA, HotpotQA, SQuADv2, GSM8K, and MMLU. These benchmarks cover

Task	Prompt template
Open-domain QA	Answer with a short factual answer (1–5 words).
MMLU	Output only the correct option letter (A–D).
GSM8K	Show reasoning and output the final numeric answer.

Table 1: Prompt templates used for each task.

three task types: (i) open-domain question answering with TriviaQA, HotpotQA, and SQuADv2; (ii) mathematical problem solving (GSM8K); and (iii) multiple-choice reasoning (MMLU). We collect statistics from several model families and sizes, including Llama 3.1 8B and Qwen 2.5 (1.5B, 7B, 14B); for additional ablations we also include Qwen 3 4B.

4.1 Prompts

For consistency across datasets within each task category, we use a single standardized prompt template per task, displayed in Table 1. We initially used various versions of prompts, attempting to enforce a JSON-formatted output for easy parsing and post-processing, but following this format proved to be too difficult for some of the Language Models we worked with. Another early version of our prompts also had more examples (since it is a few-shot prompt), but these were enforcing some undesired behaviors, so we settled with the simple one-shot approach in the end.

The full prompts are shown in Appendix A.

4.2 Features

For each example, we collect a set of internal model features, both logit-based statistics captured during generation and final-layer statistics taken after generation. Logit-based features summarize uncertainty in the model’s output probability distribution during decoding, and we take the mean and standard deviation of token entropy, the mean and minimum margin between the top two logits, the mean log-probability of generated tokens, and the average top-1 token probability across the sequence. We additionally record data about model outputs such as the answer length and whether the model produces an explicit abstention (e.g., “unanswerable”). Representation-based features are derived from internal hidden states and capture latent semantic and reasoning signals that are not directly observ-

able from output probabilities. Specifically, we extract hidden representations from both a middle and the final transformer layer over the generated answer tokens, using both the final-token representation and a mean-pooled representation across tokens. All representations are L2-normalized and truncated to 256 dimensions in an attempt to use these features for all the models we have chosen to work with. When collecting data, generation did not reliably expose consistent hidden states across decoding steps and model backends, so we collect hidden representations via a separate teacher-forced forward pass over the concatenation of the prompt and the model’s generated output, ensuring stable and model-agnostic feature extraction.

An important caveat is that at times (although quite rare in our dataset), correctness labels can occasionally be misrepresented if the Language Model outputs information using odd tokens. For example, Qwen was prone to outputting a Unicode quotation mark, which looks the same as the standard version we use, but this would not be picked up by exact-string matching. Therefore, these would be labeled as wrong in our initial codebase. We have put a lot of effort into finding edge cases such as these and fixing as many of them as possible. Another instance is when the gold labels from our initial datasets are not fully comprehensive.

We would also like to note that we initially tried to use fuzzy matching, but this was a challenging issue due to the simplistic nature of our datasets. As a simple example, if the model outputs “Nice, France” instead of “Paris, France” to the question “What is the capital of France”, it would be given a 50% correctness score when it should clearly be a 0%.

4.3 Additional Data

In addition to collecting the aforementioned features, we also collected some trivial data per prompt-response pair: LM output, output correctness (taken as an exact-match compared to dataset gold responses), gold answers, and an explicit abstention flag.

Dataset	Training Samples
TriviaQA	95,956
SquadV2	130,319
HotpotQA	90,447
GSM8K	7,473
MMLU	99,842

Table 2: Dataset distributions

5 Probe Training

5.1 Data Combinations

To train probes, we pool data collected across all benchmark datasets described above into a single training corpus. This combined dataset spans multiple task formats and difficulty levels, providing a diverse set of examples for learning generalizable confidence-related signals. Unless otherwise specified, probes are trained on the union of all available datasets, with indexed train/validation/mini-test splits used to ensure consistent evaluation while avoiding data duplication. Here, we choose a 70-15-15 split to form these sets. Mini-test refers to the subset of data we hold out from the training splits of the original datasets, which we use to make sure that the trained probes are aligned (since evaluation takes a while to run). It acts more as a validation set than a test set. Later, we use data from the test/val sets of the original datasets as our test dataset (depending on which split has ground-truth data available for our evaluations).

5.2 Initial Probes

In early iterations of our probe design, models were trained using all collected features without regard to their task specificity. This caused probes to exploit correlations rather than meaningful confidence signals, specifically relying on features like answer length and disregarding signals from the Language Model data. Because answer length varies systematically across datasets and task types, this behavior led to artificially strong in-distribution performance while harming generalization, especially since each dataset has a varying number of samples. Based on these observations, we explicitly separate task-specific metadata from generalizable confidence features in later experiments, and we report ablations that quantify the impact of excluding such signals.

5.3 Generalizable Probes

Our primary probe setting uses a generalizable feature set derived from the full collection of extracted model features by explicitly excluding task- and dataset-specific signals. In particular, while the raw dataset includes extra information such as answer length and explicit abstention indicators, these features are now omitted. The remaining features are probability-based statistics computed from output logits and internal hidden-state representations extracted from intermediate and final transformer layers. This configuration is our default probe setting and is intended to capture confidence-related signals that generalize across datasets and prompts.

5.4 Super Generalizable Probes

To further study probe transfer across model sizes and architectures, we define a super-generalizable feature setting that imposes a stricter constraint on the available signals. Starting from the generalizable feature set, we additionally exclude all representation-based features derived from internal hidden states, retaining only probability-based statistics computed from the model’s output distribution. This design choice was motivated by early experiments in which probes trained with hidden-state representations failed to generalize across models with differing internal architectures, specifically within the Qwen family, where layer counts and representation structures vary by model size. In such settings, probes were unable to reliably identify which layers contained salient confidence information, resulting in poor cross-model transfer. By restricting the feature set to architecture-agnostic, logit-based statistics, the super-generalizable setting yields probes that are explicitly model-agnostic.

5.5 Probe Architecture

We train correctness probes as lightweight MLP classifiers on one of the above possible feature sets. In implementation, each probe is a scikit-learn pipeline including median imputation and feature standardization followed by an MLPClassifier with ReLU activations and an SGD optimizer. The probe is trained to minimize cross-entropy loss (log loss) for binary correctness prediction, with L2-regularization controlled by the MLP’s alpha parameter and early stopping enabled using an internal validation split of the train-

ing set. This MLP design seems to provide enough nonlinearity to capture interactions among features while remaining stable in low-dimensional super-generalizable settings.

5.6 Training/Inference Note

While training the probes, we train using our own splits of the datasets. We have train/val/test splits which are all used during the training process to determine how the training is going, and we make sure that metrics stay close to each other across these splits to avoid over/underfitting. Once training is complete, we choose an optimal threshold for each probe through maximizing F1 performance on the validation set (we explain F1 in more detail in Section 5.9). This is the threshold used during inference moving forward. Mathematically, let us note threshold for probe i as τ_i . Then, for an example prompt x , during inference, we run inference on the Language Model with this prompt x and collect the features required for probe inference. Then, we run probe inference to obtain a scalar y . If $\text{probe}_i(\text{feats}) > \tau_i$, then we identify the model output as probably correct, and vice versa.

5.7 Results

We present some results of probe training in Table 3. In all cases, probes are trained on fully pooled datasets comprising all tasks and are evaluated on our own test splits from the same LM (taken from training splits of the benchmark datasets). Across both generalizable and super-generalizable settings, probe performance is stable across splits, with only minor differences between training, validation, and test results. This indicates that probes do not overfit to the training data and generalize reliably to unseen examples drawn from the same model’s output distribution.

Within each model, generalizable probes achieve the strongest overall performance, with high accuracy and ranking quality across all splits. Super-generalizable probes, which exclude representation-based features and rely only on probability statistics, exhibit lower performance. These results suggest that output-distribution statistics alone provide substantial information for predicting correctness, even when representation-level features are removed.

Probe	LM	Split	Acc	ROC	PR	τ
Gen.	Qwen2.5-7B	Train	0.91	0.97	0.93	–
		Val	0.90	0.97	0.93	0.42
		Mini-Test	0.90	0.97	0.93	–
Super-gen.	Qwen2.5-7B	Train	0.65	0.75	0.58	–
		Val	0.65	0.76	0.59	0.30
		Mini-Test	0.66	0.76	0.59	–
Gen.	Llama3.1-8B	Train	0.85	0.93	0.91	–
		Val	0.84	0.92	0.92	0.40
		Mini-Test	0.84	0.92	0.92	–
Super-gen.	Qwen2.5-1.5B	Train	0.75	0.84	0.68	–
		Val	0.75	0.85	0.69	0.34
		Mini-Test	0.75	0.84	0.68	–
Super-gen.	Qwen2.5-14B	Train	0.74	0.82	0.71	–
		Val	0.74	0.82	0.71	0.34
		Mini-Test	0.74	0.82	0.71	–
Super-gen.	Qwen3-4B	Train	0.80	0.88	0.82	–
		Val	0.80	0.88	0.83	0.36
		Mini-Test	0.79	0.88	0.82	–

Table 3: Probe performance across train, validation, and mini-test splits. Accuracy is computed using a decision threshold selected on validation and applied unchanged to train and mini-test. ROC and PR denote AUC-ROC and AUC-PR, respectively.

5.8 Datasets

Probes are evaluated on held-out splits of the original benchmark datasets that are never used during probe training. Probe training data are constructed exclusively from the training splits of the five benchmark sets, while evaluation is performed on untouched validation or test splits (depending on ground-truth availability). Specifically, we evaluate on the validation splits of TriviaQA, HotpotQA, SQuADv2, and MMLU, and on the test split of GSM8K. This ensures that the probes are tested on unseen samples and that we are not optimizing probes for the test sets.

5.9 Metrics

We evaluate probes as probabilistic binary classifiers that predict whether a language model’s response is correct.

First, we report thresholded classification metrics: accuracy, precision, recall, and F1 score (harmonic mean of precision and recall). Accuracy measures overall agreement with ground

LM	Dataset	Accuracy (%)	Samples
Qwen 2.5 7B	TriviaQA	20.58	11,313
	HotpotQA	10.14	7,405
	SQuAD2.0	15.67	11,873
	GSM8K	32.60	1,319
	MMLU	69.43	1,531
	Overall	18.08	33,441
Llama 3.1 8B	TriviaQA	60.80	11,313
	HotpotQA	19.68	7,405
	SQuAD2.0	21.23	11,873
	GSM8K	16.45	1,319
	MMLU	60.03	1,531
	Overall	35.86	33,441

Table 4: LM Performance across all datasets.

truth via exact string matching. Precision captures how often responses predicted as correct are actually correct, and recall measures how many truly correct responses are successfully identified by the probe. The F1 score captures the tradeoff between precision and recall and prevents the metric being skewed heavily by either value.

Second, we report ranking-based metrics: the area under the ROC curve (AUC-ROC) and the area under the precision–recall curve (AUC-PR). These metrics show how well probe scores rank correct responses above incorrect ones independent of threshold value. Since they are not sensitive to threshold choice, they provide a clearer view of whether probes capture meaningful correctness signals.

Third, we evaluate probabilistic calibration using the Brier score, which measures the mean squared error between predicted correctness probabilities and the true binary outcomes. Lower Brier scores show that probability estimates are better. In our scenario, higher Brier scores relate to overconfident underconfident probability outputs. Together, these metrics allow us to disentangle ranking quality, thresholded decision quality, and probability calibration, providing a comprehensive view of probe behavior.

5.10 LM Performance

To get a baseline idea of how our probes are performing, we must also understand how each of the Language Models perform on the datasets we are testing on. We take model accuracy as an exact string match with a "gold" ground-truth answer provided by the dataset. Results are in table 4.

5.11 Probe Results

We first evaluate the generalizable correctness probe baseline, trained to predict whether a model’s output is correct on held-out examples from the same pooled dataset. Table 5 reports results across datasets and models.

Across both Qwen 2.5 7B and Llama 3.1 8B, generalizable probes achieve non-trivial performance on several datasets, but results vary substantially across tasks. Because correctness prediction is a binary classification problem where label prevalence is determined by the base model’s accuracy, probe accuracy alone is not informative. Even a trivial majority-class predictor can achieve high accuracy when the model is frequently incorrect or correct.

Consistent with this, we observe cases where probe accuracy is high despite poor precision–recall trade-offs, as well as cases where ranking-based metrics indicate meaningful signal even when thresholded metrics collapse. These observations suggest that generalizable probes capture some correctness-related information, but that their behavior is sensitive to operating point selection. We analyze the mechanisms underlying these behaviors in Section 6.

LM	Dataset	Acc	Prec.	Rec.	F1	ROC	PR	Brier
Qwen2.5-7B	TriviaQA	89.40	72.34	78.52	75.30	95.30	80.97	0.07
	HotpotQA	91.70	56.91	74.57	64.55	95.17	67.00	0.05
	SQuAD2.0	91.90	74.50	73.46	73.97	95.60	78.43	0.06
	GSM8K	96.74	50.00	4.65	8.51	65.38	8.02	0.03
	MMLU	71.59	71.96	96.80	82.55	77.69	89.10	0.17
	Overall	90.27	68.79	73.77	70.15	93.39	74.47	0.07
Llama3.1-8B	TriviaQA	81.00	80.88	90.03	85.21	87.97	90.93	0.14
	HotpotQA	81.73	52.87	65.68	58.59	85.14	58.30	0.11
	SQuAD2.0	90.22	78.29	74.65	76.43	94.96	83.41	0.07
	GSM8K	83.09	12.50	0.46	0.88	54.91	17.87	0.16
	MMLU	64.01	64.20	90.53	75.12	77.19	84.96	0.20
	Overall	83.74	70.30	75.67	72.41	88.03	77.88	0.11

Table 5: Probe performance across evaluation datasets. ROC and PR denote AUC-ROC and AUC-PR; lower is better for Brier. Overall rows report sample-size-weighted averages across datasets.

5.12 Cross-Model Performance

We next examine cross-model transfer, applying correctness probes trained on one language model to other models of different parameter counts within the same family (Qwen 2.5 1.5B, Qwen 2.5 7B, Qwen 2.5 14B, and Qwen 3 4B). This setting isolates whether correctness signals learned by the probes are model-agnostic.

Across all configurations, we find that correctness probes do not reliably transfer across model scales. While ranking-based metrics such as AUC-ROC and AUC-PR often stay above random, thresholded metrics - precision, recall, and F1 collapse. This indicates that probes may preserve relative confidence ordering across models, but threshold values fail to work as intended across all model scales.

Importantly, probe accuracy in this setting must be interpreted relative to the base model’s correctness rate. In several cases, probes default to highly conservative or lenient decision rules that align with the majority class, giving superficially strong accuracy while not providing meaningful discrimination. Overall, these results indicate that correctness signals (but more specifically, chosen thresholds) are not robustly transferable across model scales without recalibration. A detailed analysis of the underlying failure modes is provided in Section 6.

5.13 Cross-Dataset Performance

We next evaluate cross-dataset transfer, applying correctness probes trained on one dataset to the

remaining datasets while keeping the underlying language model fixed. This setting probes whether internal correctness signals are task- and dataset-agnostic, rather than tied to a specific data distribution.

Overall, we observe limited and strongly asymmetric generalization across datasets. Most probe-dataset combinations collapse to trivial decision rules, resulting in near-zero F1 scores. However, a small number of transfer directions consistently allow probes to retain meaningful signal, characterized by non-trivial ranking performance and moderately balanced precision-recall trade-offs.

The strongest and most consistent transfer arises from probes trained on HotpotQA and SQuAD2.0, particularly when evaluated on other QA-style datasets. For both Qwen 2.5 7B and Llama 3.1 8B, HotpotQA-trained probes achieve the highest dataset-size-weighted F1 scores across evaluation datasets, while SQuAD2.0-trained probes show moderate transferability, especially to TriviaQA and HotpotQA. In these settings, AUC-ROC and AUC-PR remain substantially above random even when thresholded metrics degrade, indicating that probes capture partially transferable relative correctness information. The consistency of these patterns across both language models suggests that the observed transfer reflects shared task structure.

In contrast, probes trained on GSM8K or MMLU exhibit near-total transfer failure across

other datasets. Performance collapses to degenerate operating points, and both discrimination and calibration metrics degrade substantially. As in the cross-model setting, ranking-based metrics are often more stable than thresholded metrics, highlighting the brittleness of fixed decision thresholds under dataset shift. We analyze the causes of these behaviors in Section 6.

6 Error analysis

We now analyze the shared failure modes observed in Sections 5.10–5.12. We focus on explaining why correctness probes fail under distribution shift and how these failures differ across probe families and evaluation regimes.

6.1 Accuracy

Correctness probes predict a binary label whose prevalence is determined by the base model’s accuracy. Therefore, accuracy alone is an unreliable indicator of probe quality. A probe that always predicts incorrect achieves accuracy close to $1 - \pi$, while a probe that always predicts correct achieves accuracy close to π , where π is the model’s correctness rate on the dataset.

Many failure cases correspond to probes collapsing to these trivial baselines. As a result, high accuracy does not mean the probe has the ability to effectively predict correctness, and must be interpreted alongside precision, recall, F1, and calibration metrics.

6.2 Threshold Brittleness and Ranking–Classification Gap

A consistent pattern across both cross-model and cross-dataset settings is that ranking-based metrics degrade more slowly than thresholded classification metrics. In many settings, AUC-ROC and AUC-PR remain above random even when F1 collapses, which shows that probes retain the ability to rank “likely correct” answers higher than “likely incorrect” answers.

However, fixed decision thresholds chosen during training become misaligned under distribution shift. This threshold brittleness causes probes to snap to degenerate rules, either predicting almost all examples as incorrect or almost all as correct. As we can see, failures are driven not by complete loss of signal, but by miscalibrated

decision boundaries.

6.3 Task Mismatch and Dataset Effects

Correctness signals learned by probes are strongly shaped by task structure. Probes trained on QA datasets learn task-specific patterns, which become unreliable when transferred to out-of-distribution task datasets such as GSM8K or MMLU.

Dataset size is also a factor in these issues. Smaller datasets (such as GSM8K and MMLU) provide fewer examples for learning stable correctness representations, increasing sensitivity to noise and overfitting. However, we see that task mismatch alone is enough to cause failure even for large datasets, indicating that dataset size is not the sole driver.

6.4 Generalizable vs. Super-Generalizable Probes

Generalizable probes, which rely on a broader feature set, are more prone to collapse under distribution shift due to reliance on dataset and model-specific signals. Super-generalizable probes mitigate this issue by restricting features to transferable uncertainty-based signals, reducing failures and preserving ranking performance more consistently.

However, super-generalizable probes remain vulnerable to threshold brittleness and task mismatch. While they reduce dependence on dataset-specific cues, they do not fully resolve calibration issues or enable universal correctness prediction across tasks and models.

6.5 Summary

Across all evaluation regimes, correctness probe failures are systematic rather than incidental. Errors arise from a combination of label imbalance, threshold non-transferability, and task mismatch. While super-generalizable probes mitigate some of these issues, they do not yield a universal correctness signal. These findings highlight the need for adaptive calibration or task-aware mechanisms when deploying correctness probes under distribution shift.

7 Future Work/Improvements

Due to compute limitations, we were not able to explore some aspects of this project that we

initially intended to. One of these is mentioned earlier in the n -best generations inference framework. Additionally, we could collect substantially more data for the probes by doing multiple rollouts per prompt for these language models. Additionally, we note that the language models don't actually perform amazingly at the datasets we used, causing class imbalance in the data we use to train our probes. An alternative approach would be to use larger language models (which we did not have the compute to do). Another alternative we would have liked to attempt would be to complete some form of post-training on some of these Language Models to test whether if we inject the appropriate data into the Language Models, the probes would be able to detect that the model is more confident in its answers.

Finally, our results suggest that correctness probes may be more naturally suited for relative confidence estimation and selective prediction than for fixed-threshold binary classification under distribution shift. Across both cross-model and cross-dataset settings, ranking performance often degrades more slowly than thresholded metrics, indicating that probes preserve useful ordering information even when calibration fails. Future work could therefore focus on integrating correctness probes into abstention-based or risk-controlled decision pipelines, where predictions are deferred when confidence is low, rather than enforcing a single global threshold. Such approaches may better exploit the transferrable aspects of correctness signals while mitigating the brittleness observed under model and dataset shift.

8 Contributions of group members

All members contributed equally to all parts of the project.

9 Conclusion

This project allowed us to get some hands-on experience working LLMs and an opportunity to explore ideas presented in recent papers that express the idea that Language Models know what they know. It was interesting to get these probes working and actually quite challenging to do parts that felt trivial like deciding on the prompts to use, what information to collect from the LMs, how to actually train our probes, and such. The

data collection piece was especially challenging since we had to learn how to use teacher-forcing to collect the model internals reliably. Additionally, throughout the project, finding out why parts were going wrong was also a great learning experience. For example, in our early stages, we noticed that strong models (Llama 3.1 8B) was getting very low scores on supposedly easy datasets like MMLU. Then, doing an analysis on the results showed that our prompts were not possible to enforce (needing JSON-formatted outputs). A similar issue happened with the smaller qwen models (1.5B and 4B) outputting Chinese characters at random positions in the output sequences, and another very strange occurrence, which was outputting a slightly different double-quote (") character. This one was rather frustrating because we couldn't identify what the issue was since it looked identical, but was actually a different unicode character, and this was messing up a lot of our parsing. At this point, since we were having difficulties with multiple of the LMs we were using at the time, we completely changed the prompt format we were using.

10 AI Disclosure

- Did you use any AI assistance to complete this project (report and/or code)? If so, please also specify what AI(s) you used.
 - We used Claude and ChatGPT extensively to help with the completion and debugging at various stages of writing our code.

If you answered yes to the above question, please complete the following as well:

- **Free response:** Describe your overall experience with the AI. Did you use it to generate new text, write code, generate research ideas, check your own ideas, or rewrite text? How helpful was it? Did it just directly give you a good output, or did you have to edit it? Was its output ever obviously wrong or irrelevant?
 - We mainly used Claude and ChatGPT to help generate the initial codebase. The initial codebases were decent, but we noticed that neither LLM was fully up-to-date with the APIs and models we were using, so had to gradually adjust

those. We also needed to make very specific prompts to add which libraries and logging we wanted to use, the method we wanted to follow, etc. Specifically with Claude, we noticed the token budget was rather low, and with ChatGPT, it would continuously change pieces of code we did not ask for it to.

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A Full Prompts

In this section, we show the full few-shot prompts we used for inference throughout this paper (for all parts of our pipeline).

A.1 QA Datasets

A.1.1 Closed-Book Datasets

This is the prompt used for the TriviaQA and HotpotQA.

Answer the following question
with a short factual answer (1-5
words).

Q: Who wrote Hamlet?
A: William Shakespeare

Q: What is the capital of
France?
A: Paris

Q: Which planet is known as the
Red Planet?
A: Mars

Q: {QUESTION}
A:

A.1.2 SQuADv2

Answer the question based on
the context. If the question
cannot be answered based on
the context, respond with
"unanswerable".

Context: The Amazon rainforest
is a moist broadleaf forest in
South America. The majority of
the forest is in Brazil.
Question: Which country
contains most of the Amazon
rainforest?
Answer: Brazil

Context: Super Bowl 50 was
held on February 7, 2016 at
Levi's Stadium in Santa Clara,
California.
Question: Where was Super Bowl
50 held?
Answer: Levi's Stadium

Context: CONTEXT
Question: QUESTION
Answer: ""

A.2 MMLU

Answer the following multiple
choice question by outputting

only the letter (A, B, C, or D)
of the correct answer.

Question: What is the capital
of France?
A. London
B. Berlin
C. Paris
D. Madrid
Answer: C

Question: Who wrote Romeo and
Juliet?
A. Charles Dickens
B. William Shakespeare
C. Mark Twain
D. Jane Austen
Answer: B

{QUESTION}
Answer:

A.3 GSM8K

""Solve the following math
problem. Show your reasoning
and then provide the final
numerical answer.

Q: Janet's ducks lay 16 eggs
per day. She eats three for
breakfast every morning and
bakes muffins for her friends
every day with four. She sells
the remainder at the farmers'
market daily for \$2 per fresh
duck egg. How much in dollars
does she make every day at the
farmers' market?

A: Janet's ducks lay 16 eggs per
day. She eats 3 for breakfast,
so $16 - 3 = 13$ eggs remain. She
uses 4 for muffins, so $13 - 4 =$
9 eggs remain. She sells these
9 eggs for \$2 each, so $9 * 2 =$
\$18. The answer is 18.

Q: QUESTION A: ""

B Truth Tables

B.1 Llama 3.1 8B

For the generalizable probes, here are truth tables of our
probes for each of the datasets:

	Predicted Incorrect	Predicted Correct
Actual Incorrect	1095	7
Actual Correct	216	1

Table 6: Truth table for probe predictions on GSM8K
(Llama3.1-8B).

	Predicted Incorrect	Predicted Correct
Actual Incorrect	5095	853
Actual Correct	500	957

Table 7: Truth table for probe predictions on HotpotQA (Llama3.1-8B).

	Predicted Incorrect	Predicted Correct
Actual Incorrect	148	464
Actual Correct	87	832

Table 8: Truth table for probe predictions on MMLU (Llama3.1-8B).

	Predicted Incorrect	Predicted Correct
Actual Incorrect	8830	522
Actual Correct	639	1882

Table 9: Truth table for probe predictions on SQuAD2.0 (Llama3.1-8B).

	Predicted Incorrect	Predicted Correct
Actual Incorrect	2971	1464
Actual Correct	686	6192

Table 10: Truth table for probe predictions on TriviaQA (Llama3.1-8B).

B.2 Qwen 2.5 7B

Here, we present truth tables for the generalizable probe on Qwen 2.5 7B across datasets.

	Predicted Incorrect	Predicted Correct
Actual Incorrect	1274	2
Actual Correct	41	2

Table 11: Truth table for probe predictions on GSM8K (Qwen2.5-7B).

	Predicted Incorrect	Predicted Correct
Actual Incorrect	6230	424
Actual Correct	191	560

Table 12: Truth table for probe predictions on HotpotQA (Qwen2.5-7B).

	Predicted Incorrect	Predicted Correct
Actual Incorrect	67	401
Actual Correct	34	1029

Table 13: Truth table for probe predictions on MMLU (Qwen2.5-7B).

	Predicted Incorrect	Predicted Correct
Actual Incorrect	9544	468
Actual Correct	494	1367

Table 14: Truth table for probe predictions on SQuAD2.0 (Qwen2.5-7B).

	Predicted Incorrect	Predicted Correct
Actual Incorrect	8268	699
Actual Correct	500	1828

Table 15: Truth table for probe predictions on (Qwen2.5-7B).

C Cross-Model Results

This section reports cross-model evaluations of super-generalizable probes trained on Qwen models and evaluated on Qwen language models of different parameter scales. All metrics are computed on the same held-out evaluation splits used in Section 5.

C.1 Baseline

In this subsection, we evaluate the generalizable probe trained on Qwen 2.5 7B when applied to Qwen models of different parameter scales (Table 16).

LM	Dataset	M.Acc	P.Acc	Prec.	Rec.	F1	ROC	PR	Brier
Qwen 2.5 1.5B	TriviaQA	29.39	71.17	57.66	7.24	12.88	60.89	40.80	0.25
	HotpotQA	12.42	87.16	38.57	5.88	10.20	68.12	25.63	0.11
	SQuAD2.0	10.46	89.78	65.93	4.83	9.00	59.20	27.45	0.09
	GSM8K	2.05	96.44	4.54	3.73	4.08	67.77	3.53	0.03
	MMLU	54.87	45.13	0.00	0.00	0.00	47.91	54.59	0.53
	Overall	19.00	81.12	51.63	5.61	9.97	58.73	31.86	0.17
Qwen 3 4B	TriviaQA	38.91	61.09	0.00	0.00	0.00	50.15	38.99	0.39
	HotpotQA	15.14	84.86	0.00	0.00	0.00	50.13	15.20	0.15
	SQuAD2.0	25.20	74.79	0.00	0.00	0.00	50.00	25.21	0.25
	GSM8K	26.08	73.92	0.00	0.00	0.00	50.00	26.08	0.26
	MMLU	71.13	28.87	0.00	0.00	0.00	50.00	71.13	0.71
	Overall	29.75	70.25	0.00	0.00	0.00	0.00	29.79	0.30
Qwen 2.5 14B	TriviaQA	56.78	43.22	0.00	0.00	0.00	45.87	52.00	0.53
	HotpotQA	18.84	81.13	0.00	0.00	0.00	39.41	15.16	0.18
	SQuAD2.0	4.31	95.56	0.00	0.00	0.00	26.84	2.77	0.04
	GSM8K	30.63	68.84	0.00	0.00	0.00	63.78	38.05	0.30
	MMLU	77.20	22.80	0.00	0.00	0.00	62.60	83.49	0.76
	Overall	29.65	70.28	0.00	0.00	0.00	39.15	27.26	0.28

Table 16: Baseline results using the generalizable Qwen 2.5 7B probe on Qwen models of other sizes.

C.2 Qwen 2.5 1.5B Probes

LM	Dataset	M.Acc	P.Acc	Prec.	Rec.	F1	ROC	PR	Brier
Qwen 3 4B	TriviaQA	38.91	43.80	40.36	93.05	56.30	50.36	38.79	0.51
	HotpotQA	15.14	17.33	15.25	97.86	26.38	52.76	15.95	0.80
	SQuAD2.0	25.21	25.42	25.26	99.97	40.33	52.59	26.23	0.73
	GSM8K	26.08	26.08	26.08	100.00	41.37	50.65	26.34	0.74
	MMLU	71.13	71.13	71.13	100.00	83.13	50.00	71.13	0.28
	Overall	29.75	31.96	30.28	97.16	44.64	51.68	30.26	0.65
Qwen 2.5 7B	TriviaQA	20.84	29.06	19.69	78.03	31.44	26.77	13.88	0.62
	HotpotQA	10.17	20.34	10.60	91.90	19.00	49.94	9.69	0.65
	SQuAD2.0	15.73	21.69	16.65	99.25	28.51	64.69	20.79	0.54
	GSM8K	3.26	14.03	3.18	86.05	6.13	44.85	2.89	0.63
	MMLU	69.37	68.39	69.69	96.33	80.87	64.56	78.36	0.27
	Overall	18.19	25.72	18.23	89.79	28.91	47.81	17.93	0.58
Qwen 2.5 14B	TriviaQA	56.78	57.03	56.93	99.94	72.54	62.18	63.55	0.40
	HotpotQA	18.84	19.59	18.98	100.00	31.91	61.16	23.12	0.75
	SQuAD2.0	4.31	6.95	4.39	99.02	8.41	51.91	4.48	0.90
	GSM8K	30.63	31.99	31.02	99.75	47.33	53.94	32.41	0.67
	MMLU	77.20	77.20	77.20	100.00	87.14	50.45	77.36	0.23
	Overall	29.65	30.90	29.78	99.62	40.44	57.44	33.03	0.66

Table 17: Cross-model results using the super generalizable Qwen 2.5 1.5B probe on Qwen models of other sizes.

C.3 Qwen 3 4B Probes

LM	Dataset	M.Acc	P.Acc	Prec.	Rec.	F1	ROC	PR	Brier
Qwen 2.5 1.5B	TriviaQA	29.39	76.57	62.85	49.56	55.42	77.19	63.37	0.16
	HotpotQA	12.41	82.11	30.99	36.02	33.31	72.34	28.06	0.11
	SQuAD2.0	10.46	89.05	47.76	49.84	48.78	81.23	48.62	0.09
	GSM8K	2.05	84.53	0.00	0.00	0.00	29.88	1.40	0.09
	MMLU	54.87	43.89	27.91	1.43	2.72	44.89	49.25	0.44
	Overall	19.00	81.04	46.36	42.50	43.57	74.21	47.23	0.14
Qwen 2.5 7B	TriviaQA	20.84	82.55	66.93	32.27	43.55	87.20	61.40	0.13
	HotpotQA	10.17	87.89	28.18	12.35	17.17	83.50	28.46	0.09
	SQuAD2.0	15.73	80.70	25.49	11.78	16.11	65.86	23.34	0.14
	GSM8K	3.26	96.13	0.00	0.00	0.00	48.33	3.59	0.06
	MMLU	69.37	30.70	57.14	0.37	0.74	40.04	63.32	0.46
	Overall	18.19	81.24	40.55	17.85	24.29	75.11	38.40	0.14
Qwen 2.5 14B	TriviaQA	56.78	43.30	51.20	2.99	5.65	63.47	62.77	0.37
	HotpotQA	18.84	79.05	21.94	4.37	7.29	66.45	26.04	0.15
	SQuAD2.0	4.31	81.50	1.50	5.08	2.31	67.48	5.95	0.09
	GSM8K	30.63	75.28	90.62	21.53	34.80	82.70	70.05	0.19
	MMLU	77.20	62.25	88.04	59.14	70.75	68.61	88.49	0.32
	Overall	29.65	66.91	30.32	7.34	8.96	66.54	35.93	0.21

Table 18: Baseline results using the super generalizable Qwen 3 4B probe on Qwen models of other sizes.

C.4 Qwen 2.5 7B Probes

7B Probe results.

LM	Dataset	M.Acc	P.Acc	Prec.	Rec.	F1	ROC	PR	Brier
Qwen 2.5 1.5B	TriviaQA	29.39	60.12	40.90	80.27	54.19	73.48	56.04	0.19
	HotpotQA	12.41	58.15	17.60	64.42	27.64	66.60	21.16	0.15
	SQuAD2.0	10.46	69.98	21.82	72.38	33.53	77.71	40.93	0.11
	GSM8K	2.04	62.93	1.47	25.93	2.78	45.47	1.78	0.09
	MMLU	54.87	49.71	56.14	38.10	45.39	47.31	52.75	0.35
	Overall	19.00	62.82	28.11	69.89	38.55	71.11	40.66	0.16
Qwen 3 4B	TriviaQA	38.91	63.25	54.53	33.35	41.39	71.00	58.13	0.25
	HotpotQA	15.14	80.81	29.67	19.54	23.56	74.80	29.69	0.12
	SQuAD2.0	25.21	75.12	50.70	46.98	48.77	79.35	55.83	0.15
	GSM8K	26.08	58.76	20.06	19.48	19.76	52.93	25.53	0.20
	MMLU	71.13	49.38	76.70	41.41	53.79	54.16	74.55	0.38
	Overall	29.75	70.54	47.32	34.95	39.78	73.32	50.48	0.19
Qwen 2.5 14B	TriviaQA	56.78	67.60	69.53	76.44	72.82	68.48	66.11	0.27
	HotpotQA	18.84	61.99	29.76	74.84	42.59	70.27	28.57	0.15
	SQuAD2.0	4.31	87.59	22.23	75.20	34.31	88.61	26.14	0.06
	GSM8K	30.63	66.11	43.58	36.14	39.51	68.70	40.37	0.20
	MMLU	77.20	30.76	65.33	22.00	32.91	35.05	69.70	0.48
	Overall	29.65	71.71	42.71	71.56	49.31	74.50	42.75	0.18

Table 19: Cross-model results using the super generalizable Qwen 2.5 7B probe on Qwen models of other sizes.

C.5 Qwen 2.5 14B Probes

LM	Dataset	M.Acc	P.Acc	Prec.	Rec.	F1	ROC	PR	Brier
Qwen 2.5 1.5B	TriviaQA	29.39	70.52	22.22	0.12	0.24	76.31	59.44	0.27
	HotpotQA	12.41	87.25	3.70	0.11	0.21	75.07	26.89	0.12
	SQuAD2.0	10.46	89.53	33.33	0.08	0.16	68.60	42.47	0.10
	GSM8K	2.04	97.95	0.00	0.00	0.00	37.53	1.66	0.02
	MMLU	54.87	45.26	100	0.23	0.47	42.03	49.42	0.54
	Overall	19.00	80.90	24.75	0.10	0.21	70.20	43.47	17.67
Qwen 3 4B	TriviaQA	38.91	63.59	55.26	33.78	41.93	70.65	54.47	0.26
	HotpotQA	15.14	72.61	25.86	43.35	32.40	71.81	28.13	0.15
	SQuAD2.0	25.21	65.34	28.69	25.23	26.84	65.50	32.18	0.22
	GSM8K	26.08	52.24	31.99	73.84	44.64	58.46	28.78	0.25
	MMLU	71.13	33.18	59.65	18.73	28.51	38.45	65.15	0.46
	Overall	29.75	64.37	38.60	33.75	33.96	67.12	40.20	0.23
Qwen 2.5 7B	TriviaQA	20.84	70.54	1.40	0.59	0.83	46.07	17.70	0.22
	HotpotQA	10.17	84.88	12.01	7.70	9.39	72.71	18.39	0.10
	SQuAD2.0	15.73	78.68	18.87	10.76	13.71	79.50	30.39	0.13
	GSM8K	3.26	90.07	2.17	4.65	2.96	42.93	2.78	0.06
	MMLU	69.37	52.58	93.08	34.18	50.00	68.22	83.88	0.38
	Overall	18.19	76.55	14.18	7.48	9.63	64.73	24.80	0.16

Table 20: Cross-model results using the super generalizable Qwen 2.5 1.5B probe on Qwen models of other sizes.

D Cross-Dataset Results

In this section, we report cross-dataset evaluations of our probes. We train on one dataset and test on the remaining four. We do these experiments for Qwen 2.5 7B and Llama 3.1 8B.

D.1 Trained on TriviaQA

LM	Dataset	M.Acc	P.Acc	Prec.	Rec.	F1	ROC	PR	Brier
Qwen 2.5 7B	HotpotQA	10.14	88.04	42.78	53.26	47.45	91.61	43.62	0.07
	SQuAD2.0	15.67	89.14	69.17	55.35	61.49	90.13	65.09	0.08
	GSM8K	3.26	96.74	0.00	0.00	0.00	50.31	7.07	0.03
	MMLU	69.43	37.95	79.58	14.30	24.24	45.04	71.15	0.54
	Overall	17.47	88.01	60.91	42.13	49.94	81.09	57.37	0.14
Llama 3.1 8B	HotpotQA	19.68	73.07	37.49	55.25	44.67	74.75	41.67	0.16
	SQuAD2.0	21.23	79.24	50.66	85.20	63.54	88.85	68.96	0.13
	GSM8K	16.45	62.47	7.62	11.52	9.17	38.70	12.65	0.29
	MMLU	60.03	43.96	65.02	14.36	23.53	61.28	65.91	0.46
	Overall	20.94	71.32	56.01	44.93	49.40	70.03	54.26	0.18

Table 21: Results from testing probes trained on the TriviaQA dataset on other datasets.

D.2 Trained on HotpotQA

LM	Dataset	M.Acc	P.Acc	Prec.	Rec.	F1	ROC	PR	Brier
Qwen 2.5 7B	TriviaQA	20.58	84.04	72.48	36.21	48.30	91.61	67.65	0.12
	SQuAD2.0	15.67	86.59	57.71	54.11	55.85	86.59	56.11	0.30
	GSM8K	3.26	96.74	0.00	0.00	0.00	53.27	4.83	0.03
	MMLU	69.43	42.06	65.38	35.18	45.75	44.28	70.00	0.51
	Overall	32.23	84.87	66.05	36.67	45.83	81.67	62.55	0.16
Llama 3.1 8B	TriviaQA	60.80	77.54	82.23	80.44	81.33	84.31	87.96	0.20
	SQuAD2.0	21.23	83.27	57.66	79.85	66.97	89.70	69.39	0.10
	GSM8K	16.45	81.05	2.86	0.46	0.79	54.10	17.02	0.17
	MMLU	60.03	40.17	61.54	0.87	1.72	60.72	66.19	0.57
	Overall	41.43	63.75	56.34	56.68	56.44	81.61	63.78	0.18

Table 22: Results from testing probes trained on the HotpotQA dataset on other datasets.

D.3 Trained on SQuAD2.0

LM	Dataset	M.Acc	P.Acc	Prec.	Rec.	F1	ROC	PR	Brier
Qwen 2.5 7B	TriviaQA	20.58	80.90	71.14	12.07	20.64	89.16	60.93	0.64
	HotpotQA	10.41	89.45	44.32	15.80	23.05	89.25	38.66	0.08
	GSM8K	3.26	96.74	0.00	0.00	0.00	50.82	3.99	0.03
	MMLU	69.43	32.27	79.55	3.29	6.32	45.62	71.59	0.59
	Overall	27.78	78.98	63.09	12.63	20.57	78.92	53.74	0.47
Llama 3.1 8B	TriviaQA	60.80	65.98	78.57	60.56	68.39	74.15	79.27	0.26
	HotpotQA	19.68	74.04	32.92	30.75	31.80	60.64	30.15	0.16
	GSM8K	16.45	83.54	0.00	0.00	0.00	59.16	20.45	0.16
	MMLU	60.03	39.97	50.00	0.11	0.22	59.24	65.10	0.60
	Overall	40.63	63.39	53.31	37.16	43.25	67.81	63.11	0.30

Table 23: Results from testing probes trained on the SQuAD2.0 dataset on other datasets.

D.4 Trained on GSM8K

LM	Dataset	M.Acc	P.Acc	Prec.	Rec.	F1	ROC	PR	Brier
Qwen 2.5 7B	TriviaQA	20.58	79.52	59.32	1.50	2.93	67.17	34.74	0.19
	HotpotQA	10.14	89.70	7.14	0.13	0.26	65.32	16.19	0.10
	SQuAD2.0	15.67	84.33	0.00	0.00	0.00	48.38	15.42	0.15
	MMLU	69.43	30.57	0.00	0.00	0.00	47.59	68.31	0.66
	Overall	27.22	74.13	20.97	1.05	1.96	60.55	33.79	0.23
Llama 3.1 8B	TriviaQA	60.80	39.20	0.00	0.00	0.00	55.95	64.01	0.55
	HotpotQA	19.68	80.32	0.00	0.00	0.00	49.97	19.60	0.18
	SQuAD2.0	21.23	78.77	0.00	0.00	0.00	75.08	36.39	0.19
	MMLU	60.03	39.97	0.00	0.00	0.00	56.62	63.93	0.55
	Overall	40.92	60.71	0.00	0.00	0.00	59.96	37.41	0.45

Table 24: Results from testing probes trained on the GSM8K dataset on other datasets.

D.5 Trained on MMLU

LM	Dataset	M.Acc	P.Acc	Prec.	Rec.	F1	ROC	PR	Brier
Qwen 2.5 7B	TriviaQA	20.58	22.09	20.71	98.50	34.23	64.11	34.32	0.62
	HotpotQA	10.14	14.00	10.52	99.60	19.02	73.64	19.99	0.60
	SQuAD2.0	15.67	19.09	16.21	99.79	27.88	78.48	38.28	0.57
	GSM8K	3.26	8.72	3.37	97.67	6.52	60.36	5.12	0.50
	Overall	16.72	17.67	14.73	99.12	25.44	70.29	31.25	0.59
Llama 3.1 8B	TriviaQA	60.80	63.63	62.68	99.26	76.84	71.51	76.52	0.29
	HotpotQA	19.68	28.52	21.14	96.43	34.68	68.05	33.98	0.52
	SQuAD2.0	21.23	30.54	23.24	98.61	37.61	72.37	36.97	0.54
	GSM8K	16.45	68.46	13.82	17.51	15.45	51.51	16.25	0.21
	Overall	39.72	50.17	47.44	80.41	59.56	68.93	55.42	0.37

Table 25: Results from testing probes trained on the MMLU dataset on other datasets.